

American black bear

The **American black bear** (*Ursus americanus*), or simply **black bear**, is a species of medium-sized bear which is endemic to North America. It is the continent's smallest and most widely distributed bear species. It is an omnivore, with a diet varying greatly depending on season and location. It typically lives in largely forested areas; it will leave forests in search of food and is sometimes attracted to human communities due to the immediate availability of food.

The International Union for Conservation of Nature (IUCN) lists the American black bear as a least-concern species because of its widespread distribution and a large population, estimated to be twice that of all other bear species combined. Along with the brown bear (*Ursus arctos*), it is one of the two modern bear species not considered by the IUCN to be globally threatened with extinction.

Taxonomy

The American black bear is not closely related to the brown bear or polar bear, though all three species are found in North America; genetic studies reveal that they split from a common ancestor 5.05 million years ago (mya).^[3] American and Asian black bears are considered sister taxa and are more closely related to each other than to the other modern species of bears.^{[3][4]}

Evolution

The ancestors of American black bears and Asian black bears diverged from sun bears 4.58 mya. The American black bear then split from the Asian black bear 4.08 mya.^{[3][5]} A small, primitive bear genus called protarctos (*U. abstrusus*) is the oldest known North American fossil member of the genus *Ursus*, dated to 4.95 mya.^[6] This suggests that *U. abstrusus* may be the direct ancestor of the American black bear, which evolved in North America.^{[3][7]}

American black bear

Temporal range: Late Pliocene–Holocene



An American black bear in Manitoba's Riding Mountain National Park

Conservation status



Least Concern (IUCN 3.1)^[1]

CITES Appendix II ^[2]

Scientific classification

Kingdom:	<u>Animalia</u>
Phylum:	<u>Chordata</u>
Class:	<u>Mammalia</u>
Order:	<u>Carnivora</u>
Family:	<u>Ursidae</u>
Subfamily:	<u>Ursinae</u>
Genus:	<u><i>Ursus</i></u>

The earliest American black bear fossils, from the Early Pleistocene of Port Kennedy, Pennsylvania, greatly resemble the Asian species,^[8] though later specimens grew to sizes comparable to grizzly bears.^[9] Once described as a precursor species (*Ursus vitabilis*),^{[10][11][12]} these specimens have been synonymized with *U. americanus*.^{[7][13]} The American black bear lived during the same period as the giant and lesser short-faced bears (*Arctodus simus* and *A. pristinus*, respectively) and the Florida spectacled bear (*Tremarctos floridanus*). These tremarctine bears evolved from bears that had emigrated from Asia to the Americas 7–8 mya.^[14] The giant and lesser short-faced bears are thought to have been heavily carnivorous and the Florida spectacled bear more herbivorous,^[15] while the American black bears remained arboreal omnivores, like their Asian ancestors. From the Holocene to the present, American black bears seem to have shrunk in size,^[3] but this has been disputed because of problems with dating these fossil specimens.^[10]

The American black bear's generalist behavior allowed it to exploit a wider variety of foods and has been given as a reason why, of these three genera, it alone survived climate and vegetative changes through the last Ice Age while the other, more specialized North American predators became extinct.^[9] However, both *Arctodus* and *Tremarctos* had survived several other, previous ice ages. After these prehistoric ursids became extinct at the end of the Pleistocene, American black bears, brown bears and polar bears were the only remaining bears in North America.^[16]

Hybrids

American black bears are reproductively compatible with several other bear species and occasionally produce hybrid offspring. According to Jack Hanna's *Monkeys on the Interstate*, a bear captured in Sanford, Florida, was thought to have been the offspring of an escaped female Asian black bear and a male American black bear.^[17] In 1859, an American black bear and a Eurasian brown bear were bred together in the London Zoo, but the three cubs that were born died before they reached maturity.^[18] In *The Variation of Animals and Plants Under Domestication*, Charles Darwin noted:

In the nine-year Report it is stated that the bears had been seen in the zoological gardens to couple freely, but previously to 1848 most had rarely conceived. In the reports published since this date three species have produced young (hybrids in one case), ...^[19]

Species: ***U. americanus***

Binomial name

Ursus americanus

Pallas, 1780

Subspecies

16, see table



American black bear range^[1]

☐ Present-day range ☐ Extirpated

Synonyms

Euarctos americanus

Ursus (Euarctos) vitabilis

Gidley, 1913

A bear shot in autumn 1986 in Michigan was thought by some to be an American black bear/grizzly bear hybrid, because of its unusually large size and its proportionately larger brain case and skull. DNA testing was unable to determine whether it was a large American black bear or a grizzly bear.^[20]

Subspecies

Sixteen subspecies are traditionally recognized; however, a 2015 genetic study did not support designating some of these, such as the Florida black bear, as distinct subspecies.^[21] Listed alphabetically according to subspecific name:^{[22][23]}

American black bear subspecies

Image	Scientific name	Common name	Distribution	Description
	<i>Ursus americanus altifrontalis</i>	Olympic black bear	the Pacific Northwest coast from central British Columbia through northern California and inland to the tip of northern Idaho and British Columbia	
	<i>Ursus americanus amblyceps</i>	New Mexico black bear	Colorado, New Mexico, western Texas and the eastern half of Arizona into northern Mexico and southeastern Utah	
	<i>Ursus americanus americanus</i>	Eastern black bear	Eastern Montana to the Atlantic coast, from Alaska south and east through Canada to Maine and south to Texas. Thought to be increasing in some regions.	Common to Eastern Canada and the eastern U.S. wherever suitable habitat is found. A large-bodied subspecies; almost all specimens have black fur. May very rarely sport a white blaze on the chest.
	<i>Ursus americanus californiensis</i>	California black bear	the mountain ranges of southern California, north through the Central Valley to southern Oregon	Able to live in varied climates: found in temperate rain forest in the north and chaparral shrubland in the south. Small numbers may feature cinnamon-colored fur.
	<i>Ursus americanus carlottae</i>	Haida Gwaii black bear or Queen Charlotte Islands black bear	Haida Gwaii (formerly the Queen Charlotte Islands) and Alaska	Generally larger than its mainland counterparts with a large skull and molars and found only in a black color phase. ^[24]
	<i>Ursus americanus cinnamomum</i>	Cinnamon bear	Colorado, Idaho, western Montana and Wyoming, eastern Washington and Oregon and northeastern Utah	Has brown or reddish-brown fur, reminiscent of cinnamon.
	<i>Ursus americanus emmonsii</i>	Glacier bear or blue bear	Southeastern Alaska	Distinguished by its fur being silvery-gray with a blue luster found mostly on its flanks. ^[25]
	<i>Ursus americanus eremicus</i>	East Mexican black bear	Northeastern Mexico and U.S. borderlands with Texas.	Most often found in Big Bend National Park and the desert border with Mexico. Numbers unknown in Mexico but are presumed to be very low. Critically Endangered.

	<i>Ursus americanus floridanus</i>	<u>Florida black bear</u>	Florida, southern Georgia, Alabama and Mississippi (except the southern region)	Has a light brown nose and shiny black fur. A white blaze on the chest is common in this subspecies. An average male weighs 136 kg (300 lb).
	<i>Ursus americanus hamiltoni</i>	<u>Newfoundland black bear</u>	<u>Newfoundland</u>	Generally bigger than its mainland relatives, ranging in size from 90 to 270 kg (200 to 600 lb) and averaging 135 kg (298 lb). It has one of the longest hibernation periods of any bear in North America. ^[26] Known to favor foraging in fields of <i>Vaccinium</i> species.
	<i>Ursus americanus kermodei</i>	<u>Spirit bear or island white bear, Kermode bear</u>	the central coast of British Columbia	Approximately 10% of the population of this subspecies have white or cream-colored coats due to a recessive gene. The other 90% appear as normal-colored black bears. ^[27]
	<i>Ursus americanus luteolus</i>	<u>Louisiana black bear</u>	Eastern Texas, Louisiana and southern Mississippi. The validity of this subspecies has been repeatedly disputed. ^[1]	Has relatively long, narrow and flat skull and proportionately large molars. ^[28] Prefers <u>bottomland forests</u> and <u>bayous</u> as habitat.
	<i>Ursus americanus machetes</i>	West Mexican black bear	north-central Mexico	
	<i>Ursus americanus perniger</i>	Kenai black bear	the <u>Kenai Peninsula</u> , Alaska	Considered an "Apparently Secure Subspecies" by NatureServe. ^[29]
	<i>Ursus americanus pugnax</i>	Dall Island black bear	Dall Island in the <u>Alexander Archipelago</u> , Alaska	
	<i>Ursus americanus vancouveri</i>	Vancouver Island black bear	Vancouver Island, British Columbia	Darker and slightly bigger than the other five subspecies found in British Columbia; it is most common in the north, but appears occasionally in the southern parts of Vancouver Island.

Distribution and population

Historically, American black bears occupied the majority of North America's forested regions. Today, they are primarily limited to sparsely settled, forested areas.^[30] American black bears currently inhabit much of their original Canadian range, though they seldom occur in the southern farmlands of Alberta, Saskatchewan and Manitoba; they have been extirpated on Prince Edward Island since

1937.^[31] Surveys taken in the mid-1990s found the Canadian black bear population to be between 396,000 and 476,000 in seven provinces;^[32] this estimate excludes populations in New Brunswick, the Northwest Territories, Nova Scotia and Saskatchewan. All provinces indicated stable populations of American black bears over the 2000s.^[30]



An American black bear at Grand Teton National Park, Wyoming

The current range in the United States is constant throughout most of the Northeast and within the Appalachian Mountains almost continuously from Maine to northern Georgia, the northern Midwest, the Rocky Mountain region, the West Coast and Alaska.^[31] However, it becomes increasingly fragmented or absent in other regions. Despite this, American black bears in those areas seem to have expanded their range in recent decades, such as with sightings in Ohio,^[31] Illinois,^[33] southern Indiana,^[34] and western Nebraska.^[35] Sightings of itinerant black bears in the Driftless Area of southeastern Minnesota, northeastern Iowa, and southwestern Wisconsin are common.^{[36][37]} In 2019, biologists with the Iowa Department of Natural Resources confirmed documentation of an American black bear living year-round in woodlands near the town of Decorah in northeastern Iowa, believed to be the first instance of a resident black bear in Iowa since the 1880s.^{[38][39]}

Surveys taken from 35 states in the early 1990s indicated that American black bear populations were either stable or increasing, except in Idaho and New Mexico. The population in the United States was estimated to range between 339,000 and 465,000 in 2011,^[40] though this estimate does not include data from Alaska, Idaho, South Dakota, Texas or Wyoming, whose populations were not recorded in the survey.^[30] In California there were an estimated 25,000-35,000 black bears in 2017, making it the largest population of the species in any of the 48 contiguous United States.^{[41][42]} In 2020 there were about 1,500 bears in Great Smoky Mountains National Park, where the population density is about two per square mile.^[43] In western North Carolina, the black bear population has dramatically increased from about 3,000 in the early 2000s to over 8,000 in the 2020s.^[44]

As of 1993, known black bear populations in Mexico existed in four areas, though knowledge on the distribution of populations outside those areas has not been updated since 1959. Mexico is the only country where the species is classified as "endangered".^[30]

Habitat

Throughout their range, habitats preferred by American black bears have a few shared characteristics. They are often found in areas with relatively inaccessible terrain, thick understory vegetation and large quantities of edible material (especially masts). The adaptation to woodlands and thick vegetation in this species may have originally been because the bear evolved alongside larger, more aggressive bear species, such as the extinct giant short-faced bear and the grizzly bear, that monopolized more open habitats^[45] and the historic presence of larger predators, such as Smilodon and the American lion, that could have preyed on black bears. Although found in the largest numbers

in wild, undisturbed areas and rural regions, American black bears can adapt to surviving in some numbers in peri-urban regions, as long as they contain easily accessible foods and some vegetative coverage.^[46]

In most of the contiguous United States, American black bears today are usually found in heavily vegetated mountainous areas, from 400 to 3,000 m (1,300 to 9,800 ft) in elevation. For American black bears living in the American Southwest and Mexico, habitat usually consists of stands of chaparral and Pinyon–juniper woodlands. In this region, bears occasionally move to more open areas to feed on prickly pear cactus. At least two distinct, prime habitat types are inhabited in the Southeastern United States. American black bears in the southern Appalachian Mountains survive in predominantly oak-hickory and mixed mesophytic forests. In the coastal areas of the southeast (such as Florida, the Carolinas and Louisiana), bears inhabit a mixture of flatwoods, bays and swampy hardwood sites.

In the northeastern part of the range (the United States and Canada), prime habitat consists of a forest canopy of hardwoods such as beech, maple, birch and coniferous species. Corn crops and oak-hickory mast are also common sources of food in some sections of the northeast; small, thick swampy areas provide excellent refuge cover largely in stands of white cedar. Along the Pacific coast, redwood, Sitka spruce and hemlocks predominate as overstory cover. Within these northern forest types are early successional areas important for American black bears, such as fields of brush, wet and dry meadows, high tidelands, riparian areas and a variety of mast-producing hardwood species. The spruce-fir forest dominates much of the range of the American black bear in the Rockies. Important non-forested areas here are wet meadows, riparian areas, avalanche chutes, roadsides, burns, sidehill parks and subalpine ridgetops.

In areas where human development is relatively low, such as stretches of Canada and Alaska, American black bears tend to be found more regularly in lowland regions.^[45] In parts of eastern Canada, especially Labrador, American black bears have adapted exclusively to semi-open areas that are more typical habitat in North America for brown bears (likely due to the absence there of brown and polar bears, as well as other large carnivore species).^[46]

Description

Build

The skulls of American black bears are broad, with narrow muzzles and large jaw hinges. In Virginia, the length of adult bear skulls was found to average 26.2 to 31.7 cm (10.3 to 12.5 in).^[45] Across its range, the longest reported skull measured from 23.5 to 35 cm (9.3 to 13.8 in).^[46] Sexual dimorphism can be seen in the tendency for females to have slenderer and more pointed faces, and conversely, for larger cheek teeth in males.^[48]

Their claws are short and curved, being thickest at the base then tapering to a point, and most often black or grayish-brown in color. Claws from both hind and front legs are almost identical in length, though the curvature of foreclaws tends to be greater. The paws of the species are relatively large, with the forepaws measuring 2.1–5.3 inches long (5.3–13.5 cm) by 2.1–5.6 inches wide (5.3–14.2 cm) and